

(K)

A fluid is blown with velocity V through a horizontal porous wall that is oscillating in its plane with velocity $U \cos(\Omega t)$. The resulting fluid velocity for $0 < y < \infty$ is given by

$$\begin{aligned} v_x &= U e^{-y/\delta} \cos(\Omega t - y/\delta), \\ v_y &= V, \end{aligned}$$

(APPROXIMATELY)

where the characteristic thickness $\delta = (2\nu/\Omega)^{1/2}$ depends on the kinematic viscosity of the fluid ν .

1. Determine the trajectory and path of a fluid particle initially located at $x = 0$ and $y = 0$.
2. Obtain the streamline that intersects the point $x = 0$ and $y = 0$.
3. Compute the vorticity $\nabla \wedge \vec{v}$. If $\nabla \wedge \vec{v} = 0$, obtain also the velocity potential.
4. Calculate the rate of expansion $\nabla \cdot \vec{v}$. If $\nabla \cdot \vec{v} = 0$, obtain also the stream function.
5. Equation of the fluid line formed by the fluid particles initially located along the vertical axis $x = 0$.
6. Determine the volume flux crossing the plane $x = 0$ as a function of time $Q(t)$ along with its average value $Q_a = \int_t^{t+2\pi/\Omega} Q dt$.
7. Obtain the volume flux leaving the sphere of radius R centered at $x = 0$ and $y = 2R$.
8. Compute the circulation around a square of side L whose base lies parallel to the wall at a distance $y = L/2$. Redo the computation making use of Stokes theorem.

1) $v_x = U e^{-y/\delta} \cos(\Omega t - y/\delta)$
 $v_y = V$

$\frac{dx}{dt} = U e^{-y/\delta} \cos(\Omega t - y/\delta) = U e^{-\frac{y}{\delta} t} \cos[(\Omega - \frac{V}{\delta})t - y_0/\delta]$
 $\frac{dy}{dt} = V \rightarrow y = Vt + y_0$
 $x - x_0 = \int_0^t U e^{-\frac{y}{\delta} t} \cos[(\Omega - \frac{V}{\delta})t - y_0/\delta] dt =$

PATHS
 $x = U e^{-\frac{y}{\delta}} \left\{ -\frac{V}{\delta} \cos[(\Omega - \frac{V}{\delta})\frac{y}{V}] + (\Omega - \frac{V}{\delta}) \sin[(\Omega - \frac{V}{\delta})\frac{y}{V}] \right\} / \left[\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2 \right] + \frac{U V/\delta}{\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2}$
 $= -\frac{U e^{-\frac{y}{\delta}}}{\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2} \left\{ \frac{V}{\delta} \cos[(\Omega - \frac{V}{\delta})\frac{y}{V}] - (\Omega - \frac{V}{\delta}) \sin[(\Omega - \frac{V}{\delta})\frac{y}{V}] \right\} + \frac{U V/\delta}{\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2}$

2) $\frac{dx}{U e^{-y/\delta} \cos(\Omega t - y/\delta)} = \frac{dy}{V} \Rightarrow \int dx = U \int_0^y e^{-y/\delta} \cos(\Omega t - y/\delta) dy = -\frac{U \delta}{2} e^{-y/\delta} [\cos(\Omega t - y/\delta) + \sin(\Omega t - y/\delta)] + \frac{U \delta}{2} [\cos(\Omega t) + \sin(\Omega t)]$

3) $\nabla \wedge \vec{v} = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \vec{e}_z = -U e^{-y/\delta} [-\cos(\Omega t - y/\delta) + \sin(\Omega t - y/\delta)] \neq 0$

4) $\nabla \cdot \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0$, $v_x = U e^{-y/\delta} \cos(\Omega t - y/\delta) = \frac{\partial \Psi}{\partial y} \Rightarrow \Psi = U \int e^{-y/\delta} \cos(\Omega t - y/\delta) dy$
 $v_y = V = -\frac{\partial \Psi}{\partial x} \Rightarrow \Psi = -Vx + f(y) = -Vx - \frac{U \delta}{2} e^{-y/\delta} [\cos(\Omega t - y/\delta) + \sin(\Omega t - y/\delta)] + \Psi_0(t)$

5) $y_0 = \lambda, x_0 = 0$ $y = Vt + \lambda$
 $x = -\frac{U e^{-\frac{y}{\delta}}}{\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2} \left\{ \frac{V}{\delta} \cos[(\Omega - \frac{V}{\delta})\frac{y}{V}] - (\Omega - \frac{V}{\delta}) \sin[(\Omega - \frac{V}{\delta})\frac{y}{V}] \right\} + \frac{U V/\delta}{\left(\frac{V}{\delta}\right)^2 + (\Omega - \frac{V}{\delta})^2}$

6) $Q = \int \vec{v} \cdot \vec{n} d\sigma = \int_0^{\infty} v_x dy = U \int_0^{\infty} e^{-y/\delta} \cos(\Omega t - y/\delta) dy = \frac{U \delta}{2} [\cos(\Omega t) + \sin(\Omega t)]$, $\langle Q_a = \int_t^{t+2\pi/\Omega} Q dt = 0 \rangle$

7) $\oint \vec{v} \cdot \vec{n} d\sigma = \int_V \nabla \cdot \vec{v} dV = 0$

8) $\Gamma = \oint \vec{v} \cdot d\vec{l} = \int_0^L \int_{-L/2}^{L/2} \int_{-L/2}^{L/2} \nabla \cdot \vec{v} dV = 0$
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